



A tiger shark swims over a sandy seabed

SHARK ATTACKS ON HUMANS ARE DECLINING GLOBALLY. SO WHY ARE THEY INCREASING IN AUSTRALIA?

A rise in cases over the past 30 years of sharks biting humans in Australia has prompted scientists to seek a solution

WORDS BY RYAN KEANE

WHEN JUSTINE BARWICK WENT ON A TRIP IN 2018 TO THE WHITSUNDAY ISLANDS, an archipelago located off Australia's north-eastern coast, she expected a relaxing holiday. One day, alongside her husband and some friends, she hopped on a yacht and sailed into the calm, cerulean expanse of the ocean. A seasoned swimmer, Barwick decided to jump into the water once they were out far enough. But shortly after she did this, the sense of tranquillity she'd been feeling quickly disappeared.

When she resurfaced moments later, something slammed into her with an almighty force. Straight away, she clocked that it was a shark that had started to attack her. Barwick somehow managed to turn and push the animal away from her, before swimming back towards the stern of the yacht. But it was once she started to ascend the boat that the severity of the attack made itself clear.

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"I remember trying to get up the swim ladder and my right leg wouldn't work," the Tasmanian resident tells *BBC Wildlife*. "I was very confused. I looked down and all of the inside of my thigh, most of the front and all of the back of it was just gone. I could actually see my thigh bone. I remember thinking 'this is pretty big' and then I saw the spurts of blood."



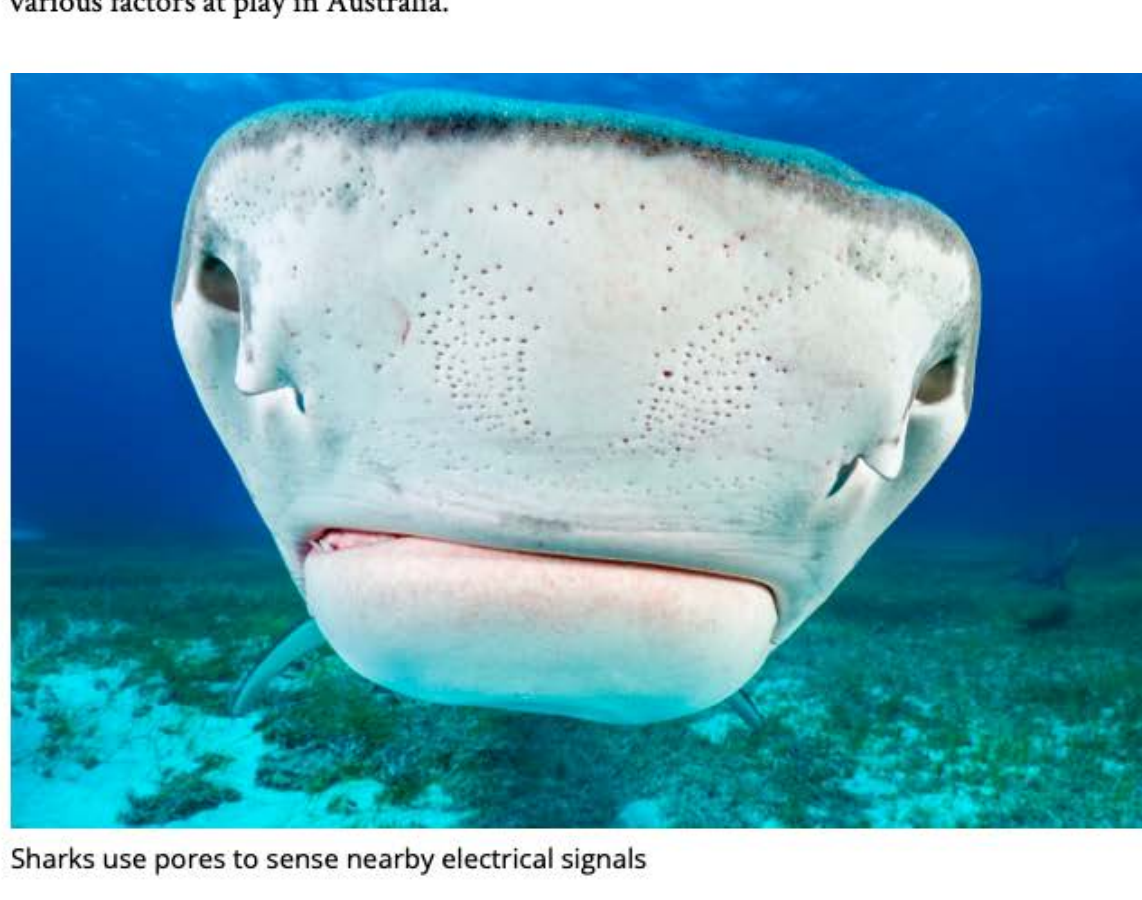
It turned out the shark had severed Barwick's femoral artery when it bit her. Upon being pulled back on to the boat by her horrified peers, she watched as they began to try and help her. Her husband instinctively grabbed a beach towel and applied pressure to the injury site. It took two to fully cover the bitten thigh. Applying layers upon layers of bandage was the next step of treatment. Due to logistics and being in such a remote location, it took almost two hours before a rescue helicopter was able to make it to the yacht. Barwick was then flown to the hospital where doctors thought she might not make it.

Thanks to the quick thinking of her husband and friends, Barwick *did* make it. Skin grafted from her stomach allowed for the reconstruction of her thigh. Yet despite being able to walk again, she now needs a walking stick, fatigue has become a familiar feeling for her, and she even had to learn to talk again.

BARWICK'S ENCOUNTER WITH WHAT WAS LIKELY A TIGER SHARK is part of a wider trend in Australia. Official statistics show that in the last three decades, cases of sharks biting humans have been increasing. In the 1990s, there were around 8–10 instances per year but, since the 2010s, the yearly case average has been in the mid-20s.

It's important to note that shark attacks in Australia are still incredibly rare despite this rise. A person's chances of being killed by, say, a great white or bull shark in the country are about one in eight million. In many cases, these aquatic dwellers might mistake our shiny wristwatches or long wetsuits for fish or seals. Since sharks lack hands, their mouths are often their most effective tool at exploring what the new stimulus in their environment is. And humans are far more dangerous to sharks than the other way around, with more than 100 million of them killed by us annually. We use their fins in soup, their meat for commercial trade, and they often get fatally entangled in fishing gear by accident.

Still, this rise in shark attacks in Australia, however small, does present questions for scientists as to why these dangerous interactions are growing in this part of the world. Jodie Rummer, a marine biologist at James Cook University in Queensland, says there are various factors at play in Australia.



Sharks use pores to sense nearby electrical signals

"What we are seeing is likely a combination of more people in the water, improved reporting [of bites when they occur] and changing ocean conditions," she says. Notably, these changing conditions relate to climate change. As coastal waters with humans pottering about in them warm more intensely, sharks are spending more time in them too, as are some of the fish they prey upon.

As Rummer also points out, marine heatwaves mess with how sharks require food and identify potential prey. "Warm water increases a shark's metabolic demand, and we know that when our metabolism is high we require more food. They're not necessarily using vision as much as we use vision to look for food. They use electroreception. They've got pores all over the front of their bodies, which can sense electrical signals in their environment."

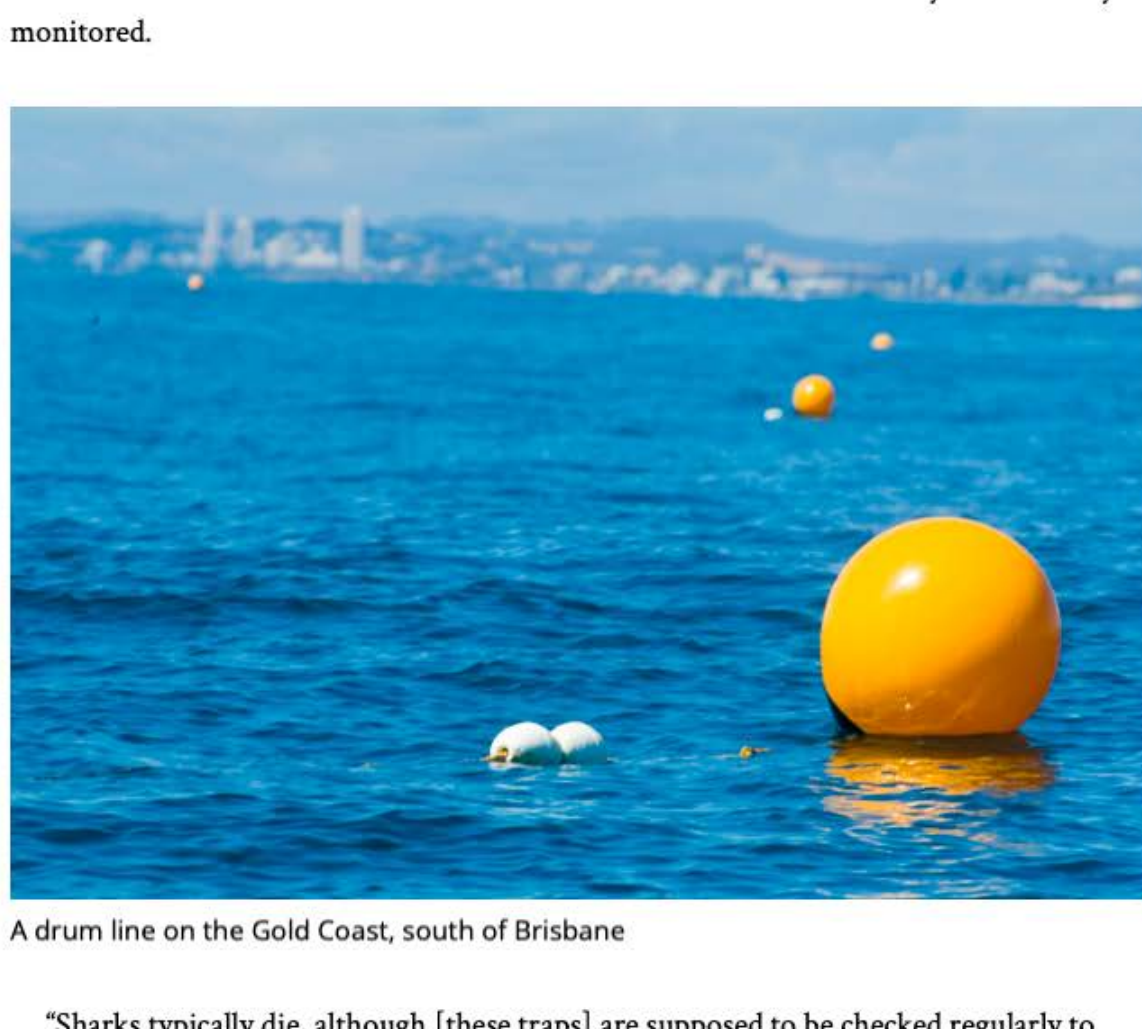
When the oceans are hotter and more acidic, water quality fades and sharks consequently rely more on their electroreception. With poorer visibility, is it any surprise that some of these sea dwellers might mistake the electrical signals from a human's heartbeat to that of a seal's or stingray's?

VARIOUS SOLUTIONS TO THE SMALL INCREASE OF ATTACKS HAVE BEEN TOUTED, with some ideas more popular in academic circles than others. Former Australian prime minister Tony Abbott received some backlash in June 2026 after he suggested that culling sharks could be key to reducing their biting of humans.

Nathan Hart, a sensory biologist at Macquarie University in Sydney, is one of the scientists unconvinced by the politician's proposal. "There's little evidence that culling reduces the risk of bites by dangerous sharks," he says. "Species such as [great] white sharks that are responsible for many of the fatal bites are highly migratory, and move up and down the coast and beyond. Even if you killed a number [of them] in an area, you can't stop others moving in from elsewhere."

For Hart, a multi-layered approach to monitoring, rather than reducing overall population, serves as a better solution. "Drones are probably the best way to manage risk at patrolled beaches, especially close to the shore and in clear water," he suggests, with these devices spotting sharks close to the surface. Then, beachgoers can be given notice of their presence as part of Australia's SharkSmart programme, especially through a mobile app offering live alerts about an animal's whereabouts.

There's also the option of drum lines, which are effectively traps anchored to the seabed that are designed to snag dangerous sharks with bait-hooks. But Hart says the more traditional forms of these tools can be harmful to marine life as they're not closely monitored.



A drum line on the Gold Coast, south of Brisbane

"Sharks typically die, although [these traps] are supposed to be checked regularly to release non-target animals," says Hart, "Newer 'smart' drum lines send an alert to an operator when an animal gets caught, and a team goes out to tag and release the shark further away from shore. This has a much better survival rate and improves what we can learn about shark movement and behaviour." It's worth adding that operators are typically said to arrive to the scene 30 minutes after an alert from a 'smart' drum line is issued.

EXPERTS HAVE EVEN BEEN WORKING ON CREATING MORE BITE-RESISTANT wetsuits for swimmers and surfers. Charlie Huveneers is a scientist at Flinders University in Adelaide. He maintains that despite chain mail-style suits being protective particularly for divers, their inflexible material often makes them too tricky to wear for more casual sea-related activities. He proposes the introduction of more materials such as "ultra-high, molecular-weight polyethylene" for wetsuits, as they offer a nice mix of flexibility and safety.

"We've tested four bite-resistant materials. While there were small differences between them, they all reduced critical damage from bites compared to standard neoprene." Neoprene, a thinner fabric, currently makes up more than 80 per cent of wetsuit material worldwide. With more surveillance, 'smarter' and less intrusive drum line traps, and better designed fabrics for swimmers and divers, marine experts such as Hart and Huveneers hope a reduction of shark attacks in Australia can occur, even though they're still incredibly rare.

Barwick, who survived her 2018 attack in the Whitsundays, can't speak scientifically on the solutions to such encounters. But she can speak, remarkably, about how her perception of these aquatic animals remains unaffected and how she views life after the incident.

"It's nobody's fault," she says. "It's not the shark's fault, it's not the ocean's fault. The work that I do with my physiotherapists keeps me physically able and enjoying life. The work that I do with my trauma psychologist keeps me mentally well. And, of course, being surrounded by people who truly want the best for me is just magic."

ABOUT THE AUTHOR



Ryan Keane is a London-based journalist who specialises in wildlife and the environment. He also works for BBC Radio 4 and the World Service. When he's not reporting, Ryan can be found relaxing with a coffee or planning his next adventure.

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